

Bakht Zada, PhD

Research Scientist

Synthetic Biology • Metabolic Engineering • Industrial Biotechnology • CRISPR Genome Engineering

Professional Summary

Research scientist with experience in biotechnology across academic and collaborative research environments. Experienced in leading independent research projects from concept to publication, mentoring undergraduate and graduate students, and contributing effectively to multidisciplinary teams. Author of peer-reviewed scientific publications with a strong commitment to high-quality research, scientific communication, and collaborative problem-solving. Focused on translating scientific discoveries into practical solutions for industrial biotechnology and sustainable biomanufacturing while advancing innovative research that addresses emerging scientific and industrial challenges.

Research Experience

07/2023 — 08/2025

Research Scholar
University of California, Davis, USA
Department of Biological and Agricultural Engineering

05/2021 — 04/2023

Senior Researcher
Gyeongsang National University, Jinju, South Korea
Department of Biochemistry

Education

09/2014 — 02/2019

Ph.D. in Applied Life Science
Gyeongsang National University, South Korea

09/2010 — 10/2012

M.S. in Biochemistry/Molecular Biology
Quaid-I-Azam University, Pakistan

09/2004 — 09/2008

B.S. in Biochemistry
University of Peshawar, Pakistan

Teaching Experience

09/2025 — Present

Assistant Professor
Government College Daggar Buner, Higher Education Department
Government of KP, Pakistan

03/2019 — 04/2021

Assistant Professor

Government College Daggar Buner, Higher Education Department
Government of KP, Pakistan

11/2012 — 08/2014

Lecturer

Government College Daggar Buner, Higher Education Department
Government of KP, Pakistan

Publications

- Zada B, Shapouri H, Wang J, Sharma S, Kasuga T, Valbuena N, Tchelet R, Emalfarb M, Fan Z. Engineering *Thermothelomyces heterothallica* for Cellobionate Production. Submitted to Bioresources and Bioprocessing.
- Zada B, Kasuga T, Fan Z. Metabolic engineering of *Escherichia coli* for Glucaric Acid Production. Manuscript in progress.
- Zada B, Joo S, Wang C, Tseten T, Jeong S-H, Seo H, Sohn J-H, Kim K-J, Kim S-W. 2020. Metabolic engineering of *Escherichia coli* for production of non-natural acetins from glycerol. **Green Chemistry** 22:7788-7802. (IF: 10.1).
<https://doi.org/10.1039/D0GC02395G>
- Zada B., Wang C., Park J. B., Jeong S. H., Park J. E., Singh H. B., Kim S-W. (2018). Metabolic engineering of *Escherichia coli* for production of mixed isoprenoid alcohols and their derivatives. **Biotechnology for Biofuels** 11: 210. (IF: 5.4).
<https://doi.org/10.1186/s13068-018-1210-0>
- Wang C, Zada B, Wei G, Kim S-W (2017a). Metabolic engineering and synthetic biology approaches driving isoprenoid production in *Escherichia coli*. **Bioresource Technology** 241:430–438 (IF: 5.8).
<https://doi.org/10.1016/j.biortech.2017.05.168>
- Tian, H.; Zada, B.; Singh, B.H.; Wang, C.; Kim, S.-W. Chapter 13 - Synthetic Biology Approaches for the Production of Isoprenoids in *Escherichia coli*. In Current Developments in Biotechnology and Bioengineering, Singh, S.P., Pandey, A., Du, G., Kumar, S., Eds.; **Elsevier**: 2019; pp. 311-329.
<https://doi.org/10.1016/B978-0-444-64085-7.00013-7>
- Zada, B.; Kwon, M.; Kim, S.-W. Current Trends in Acetins Production: Green versus Non-Green Synthesis. **Molecules** 2022, 27, 2255 (IF: 4.4).
<https://doi.org/10.3390/molecules27072255>

Patents

- Method for biologically producing acetin compounds, **U.S. Patent No. 12,275,977** (granted 15 April 2025); PCT WO 2020/085879.
<https://patents.google.com/patent/US12275977B2>
- 터펜 알코올 또는 그 유도체의 제조 방법" (Publication number: 10-2016-0074347, Publication date: 06.15.2016)
<http://synbilab.com/index.php?gt=publication/publication03>

Technical Skills

- **Genome Engineering & Synthetic Biology**
CRISPR/Cas9 genome editing (Conventional and RNP-based), recombineering

- Metabolic engineering, pathway optimization, and microbial chassis development
- **Molecular Biology & Genetic Engineering**
Advanced DNA assembly (Golden Gate, Gibson Assembly, HiFi), PCR, primer design
Plasmid construction, genomic DNA preparation, and site-directed mutagenesis
- **Protein Engineering & Biocatalysis**
Directed evolution, heterologous protein expression, purification, and enzyme characterization
Enzyme activity assays and protein analysis
- **Fermentation & Bioprocess Development**
Microbial fermentation, bioreactor operation, process optimization, and scale-up
Upstream bioprocess development and process monitoring
- **Analytical Chemistry & Multi-Omics**
HPLC, LC-MS/MS, GC-MS, GC-FID, NMR
Metabolomics, transcriptomics, proteomics, and metabolic flux analysis
- **Biochemical & Molecular Analysis**
SDS-PAGE, Western blotting, Bradford assay, spectroscopy, quantitative biochemical assays
- **High-Throughput Screening & Laboratory Management**
High-throughput screening workflows
Microbial strain libraries, plasmid repositories, and laboratory inventory management
- **Scientific Communication & Leadership**
Scientific writing, manuscript and grant preparation
Technical presentations, interdisciplinary collaboration, and mentoring graduate and undergraduate researchers
- **Bioinformatics & Scientific Software**
SnapGene, Benchling, BLAST, Vector NTI, ImageJ, GraphPad Prism, OriginLab
Microsoft Office Suite

Awards and Achievements

- Scientific Leadership and Management Skill Certificate, University of California, CA 95616, US, 2025
- Pioneer **Research Award** for Outstanding Research (2018), Division of Applied Life Science (BK21 plus Program), PMBBRC, Gyeongsang National University, Jinju 52828, South Korea.
- Pioneer **Research Award** for Outstanding Research (2017), Division of Applied Life Science (BK21 Plus Program), PMBBRC, Gyeongsang National University, Jinju 52828, South Korea
- BK 21 plus **scholarship** (2014), Gyeongsang National University, Jinju 52828, South Korea